



YISHUN INNOVA JUNIOR COLLEGE  
JC 1 PROMOTIONAL EXAMINATION  
**Higher 2**

CANDIDATE  
NAME

CG

INDEX NO

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**COMPUTING**

Paper 1 Written

**9569/01**

**3 Oct 2025**

**1 hour 10 minutes**

Additional Materials:      4-page Answer booklet

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**READ THESE INSTRUCTIONS FIRST**

An answer booklet will be provided with the question paper.  
You should follow the instructions on the front cover of the answer booklet.  
If you need additional answer paper ask the invigilator for a continuation booklet.

Answer **all** questions.  
Approved calculators are allowed.

The number of marks is given in brackets [ ] at the end of each question or part question.  
The total number of marks for this paper is **40**.

1 You are provided with the following array, a:

| Index | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|-------|----|----|----|----|----|----|----|----|
| a     | 25 | 54 | 33 | 65 | 73 | 87 | 80 | 82 |

- (a) Describe how the Insertion Sort algorithm will sort this array, showing the steps leading to the final sorted result. [4]
- (b) Compare the time efficiency of Insertion Sort and Merge Sort. Which algorithm is more suitable for sorting large unsorted arrays, and why? [2]

2 A hash table of size 10 uses **linear probing** to resolve collisions.

It applies a simple hash function that sums all the digits of a given key, then takes the result modulo 10 to determine the index.

For example, the key 2352 has digits that sum to 12. Applying the modulo function,  $12 \% 10 = 2$ , so the key is mapped to index 2.

After inserting six elements using this method, the resulting state of the hash table is shown on the right.

| Index | key  |
|-------|------|
| 0     | 6201 |
| 1     |      |
| 2     | 2352 |
| 3     | 3857 |
| 4     | 8131 |
| 5     |      |
| 6     | 5731 |
| 7     |      |
| 8     |      |
| 9     | 4249 |

- (a) Using the given hash function, describe how the key 2725 will be inserted into the hash table. Include the description of how collisions will be handled, if any. [2]
- (b) State, with justification, a possible order in which the keys can be inserted to arrive at the current state of the hash table (excluding the key 2725). [3]

**3** A game controller uses a 4-byte binary data packet to send its input to a game console.

The first three bytes (24 bits) are used to represent the X and Y positions of a pointer:

- The first 12 bits represent the X position.
- The next 12 bits represent the Y position.

The fourth byte (8 bits) represents the state of the 8 buttons on the controller:

A, B, X, Y, R, +, Home, C.

For example,

10100100

The above fourth byte corresponds to pressing the A, X, and the + buttons. The 0 indicates that the other buttons are not pressed.

A following data packet, in hexadecimal, has been sent to the game console:

3C A7 9D 42

- (a) Convert all 4 bytes of the data packet into binary digits. [2]
- (b) State the 12-bit X-axis and Y-axis positions in binary. [2]
- (c) Convert the two binary values found in (b) into decimal numbers. [2]
- (d) State the buttons which are pressed. [2]

- 4 A holiday company has some villas that customers can hire. They store the customers' and bookings' information in a relational database. Three tables present in the database are CUSTOMER, VILLA and BOOKING.

The data for the CUSTOMER table are:

customer ID, name, email

The data for the VILLA table are:

villa ID, villa name, cost

The data for the BOOKING table are:

booking ID, customer ID, villa ID, start date, number of days

- (a) Draw an Entity-Relationship (E-R) diagram to show the relationship between the customers and the bookings of the villas. [2]

- (b) A table description can be expressed as:

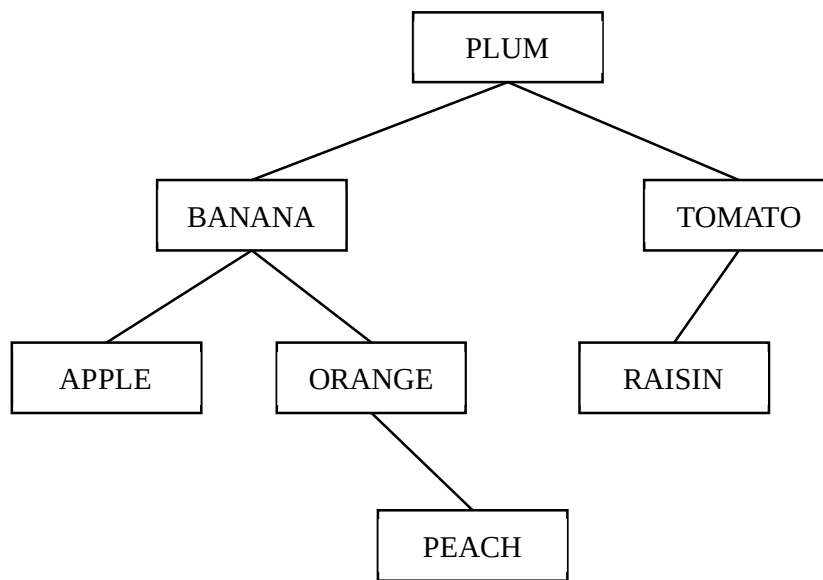
TableName (Attribute1, Attribute2, Attribute3, ...)

The primary key is indicated by underlining one or more attributes.

Foreign keys are indicated using a dashed underline.

Write the table descriptions for all the tables used in the database. [3]

5 Consider the following binary search tree (BST):



- (a) List the nodes, in order, that are visited for a post-order traversal. [2]
- (b) Without performing the in-order traversal, state the property exhibited by the order of the nodes visited. [1]
- (c) Describe an algorithm to perform a binary tree search for an item. The output should state whether or not the item is present in the BST. [5]

- 6 The employees in a newly established company work using standalone personal computers and share their files using removable storage drives.

(a) State two disadvantages of this method of sharing data files. [2]

The company is considering to set up a local area network (LAN) with a file server for the employees.

(b) Besides file sharing, suggest two additional functions that can be provided within the LAN. [2]

The company is also considering the use of an the online file service offered by a cloud services provider.

(c) State two advantages of using the services offered by the cloud services provider. [2]

(d) State an additional service which the company may consider and describe one of the benefits to the company. [2]

**END OF PAPER**